

CO₂ desorption from a packed bed worksheet

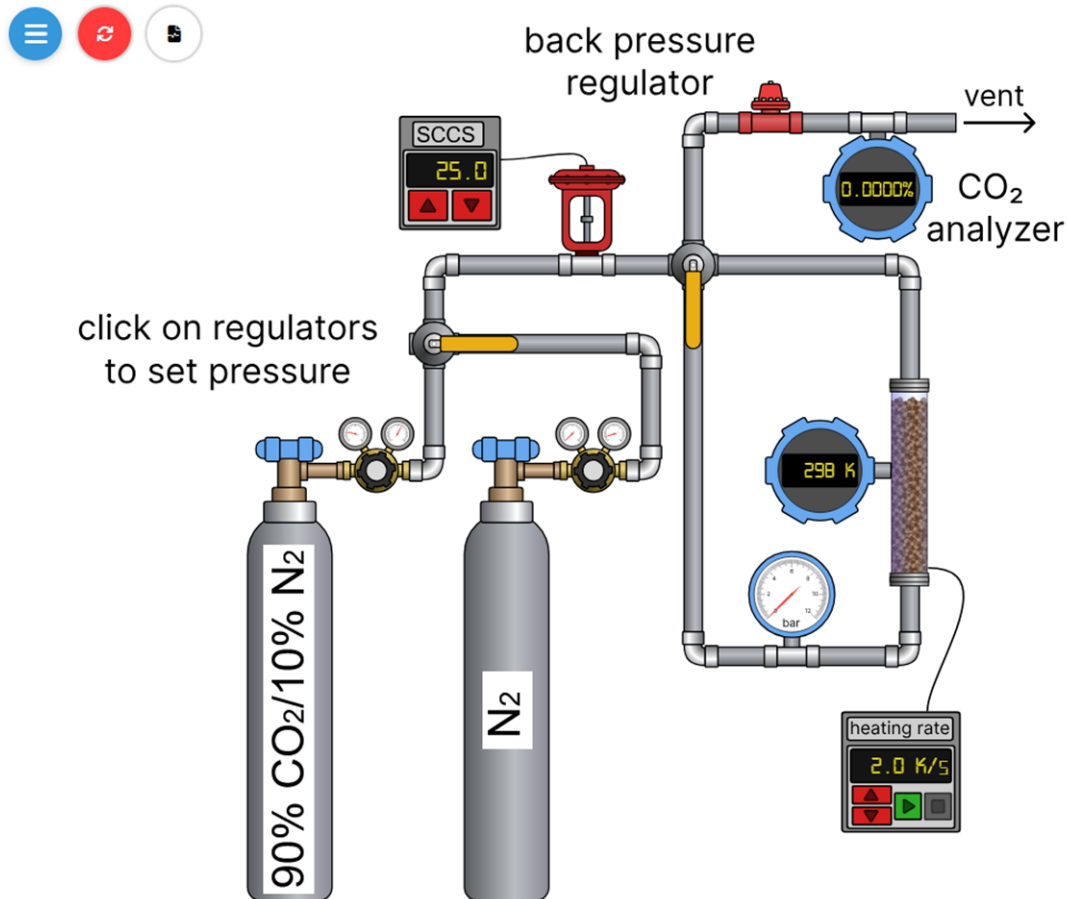
Name(s) _____

The goal of this experiment is to use an unsteady-state material balance on CO₂ to determine how much CO₂ adsorbed on a zeolite and how the desorption behavior changes with pressure and heating rate. Outlet CO₂ concentration during temperature-programmed desorption (TPD) is converted into molar flow rate of CO₂. The amount desorbed is determined from the desorption curve, and peak-temperature shifts with heating rate are used to estimate the heat of desorption.

Student Learning Objectives

1. Apply an unsteady-state mole balance to CO₂ desorption from a packed bed.
2. Convert outlet CO₂ mole fraction and total flow rate into a CO₂ molar desorption rate.
3. Integrate desorption-rate data to determine the amount of CO₂ adsorbed.
4. Compare TPD curves at different adsorption pressures and heating rates.
5. Estimate the heat of desorption from peak-temperature data.

Equipment



- A small bed filled with 1.0 g of zeolite. The bed has a temperature sensor.
- A mass flow controller that controls and measures the mass flow rate into the bed.

- A pressure gauge near bed inlet.
- A back-pressure regulator to control pressure in the bed.
- Two gas cylinders: 90 mol% CO₂/10% N₂ and N₂.
- A switching valve allows flow from one of the two gas cylinders to be fed to the bed.
- A switching valve allows flow to the zeolite bed or to bypass the bed and flow directly to the vent.
- A CO₂ gas analyzer to continuously measure the CO₂ concentration at the outlet.

For accurate TPD measurements, small zeolite particles are used to minimize mass transfer limitations. A short bed and a high N₂ flow rate allow CO₂ to be quickly removed from the bed and minimize re-adsorption. However, as the N₂ flow rate increases, N₂ dilutes the desorbing CO₂ and this reduces the signal at the CO₂ detector, reducing accuracy. If the heating rate is too high, temperature gradients in the bed would make the measurements less accurate.

Notation

$y_{CO_2,out}$ = mole fraction CO₂ leaving zeolite bed

P_{gauge} = system gauge pressure (bar)

P = system pressure absolute (bar)

v_{N_2} = N₂ volumetric flow rate (SCCS, standard cm³/s)

$P_{N_2,gauge}$ = N₂ gauge pressure during desorption (bar)

P_{N_2} = N₂ absolute pressure during desorption (bar)

T = temperature (K)

t = time (s)

P_{CO_2} = partial pressure CO₂ (bar)

β = heating rate (K/s)

T_P = peak temperature (K)

$r_{CO_2,des}$ = rate of CO₂ desorption (mol/s)

$F_{CO_2,out}$ = molar flow rate of CO₂ out of zeolite bed (mol/s)

F_{N_2} = molar flow rate of N₂ during TPD (mol/s)

$r_{CO_2,des,max}$ = maximum rate of CO₂ desorption (mol/s)

ΔH_{ads} = heat of adsorption (kJ/mol)

E_{DES} = activation energy for desorption (kJ/mol)

Questions to answer before starting the experiment

1. Sketch the expected shape of the CO₂ concentration versus temperature at the bed outlet when the reactor bed is heated.
2. At a higher heating rate, does the maximum rate of desorption appear at a lower or higher temperature? Why?

Experiment #1 Effect of CO₂ loading on TPD peak temperature

During TPD, N₂ flows through the zeolite bed and carries desorbed CO₂ to the CO₂ analyzer. Because the bed is small, not much gas accumulates in the bed so the rate of desorption $r_{CO_2,des}$ is equal to the molar flow rate of CO₂ out of the bed $F_{CO_2,out}$:

$$r_{CO_2,des} = F_{CO_2,out} = y_{CO_2,out} F_{N_2}$$

where the total flow rate out is essentially equal to the molar flow rate of N₂ into the bed F_{N_2} because the CO₂ concentration in the exit stream from the bed is small.

For first-order desorption, the TPD peak temperature is independent of initial surface coverage, assuming negligible re-adsorption and uniform binding energy. If the peak temperature shifts significantly with loading, that suggests coverage-dependent desorption energy, re-adsorption/mass-transfer effects, or desorption that is not first order.

Verify that CO₂ desorption is first order in CO₂ coverage by measuring the peak temperature T_p , which is the temperature where the desorption rate is a maximum during TPD. Carry out adsorption at low, medium, and high pressure. Obtain the amount adsorbed by integrating the desorption rate versus time curve.

Questions to answer before starting experiment

1. Why is the CO₂ peak area more important than the peak height for determining adsorption capacity?
2. If the N₂ flow rate is doubled, what happens to the measured CO₂ mole fraction at the analyzer? Does the total amount of CO₂ desorbed necessarily change?

Experimental procedure

1. Click on the blue valve at the top of the 90 mol% CO₂/10 mol% N₂ tank to open it. Then turn the regulator valve clockwise to select a pressure. Zoom with the scroll wheel to make pressure

adjustment easier. Use a gauge pressure $P_{N_2,gauge}$ of 0.2 bar and record that in Table 1. Convert it to absolute pressure and record that value P in Table 1.

2. Select a volumetric flow rate (SCCS, standard cm³/s) by clicking on the black triangles on the flow controller. The CO₂/N₂ flow goes directly to the vent. When a mouse moves over one of the orange valve handles, the CO₂ gas flow path is displayed in green. Lines with no flow are black.
3. Click on the switching valve after the flow meter to direct flow to the zeolite bed.
4. When the CO₂ analyzer reads 90%, the bed is saturated at that CO₂ pressure.
5. Carry out TPD by clicking the blue valve on the N₂ tank. Set the N₂ tank pressure $P_{N_2,gauge}$ to 0.2 bar gauge pressure. Click on the orange valve that switches the feed gas from the CO₂/N₂ mixture to N₂. When a mouse moves over one of the orange valve handles, the N₂ gas flow path is displayed in light blue.
6. When the CO₂ is flushed out of the system and the CO₂ analyzer reads zero, select the record data button (the button to the right of the red reset button) and click on the green start button to start recording data.
7. Select a heating rate using the red buttons and record the heating rate β in Table 1. Click the green button on the heating rate controller to ramp the temperature.
8. The CO₂ analyzer will show an increase in CO₂ concentration and then a decrease at higher temperature. When the CO₂ concentration decreases to zero, stop heating by clicking on the red button below the heating rate.
9. Click on the red stop button in the record data box. Then click on the download recorded data button to the right of the red stop button.
10. Download the CSV file with the recorded data and convert it to a spreadsheet. Record the peak temperature from the spreadsheet in Table 1.
11. Use the spreadsheet to integrate the area under curve of CO₂ desorption rate $r_{CO_2,des}$ versus time t to determine the amount desorbed, which is equal to the amount adsorbed. Record the value in Table 1

$$n_{CO_2,ads} = \int r_{CO_2,des} dt = \int y_{CO_2,out} F_{N_2} dt = \int y_{CO_2,out} F_{N_2} dT / \beta$$

12. Repeat these measurements for a medium and a high pressure and record the data in Table 1.

Table 1 Effect of CO₂ loading on peak temperature $P_{N_2,gauge} = 0.2$ bar $P_{N_2} = 1.2$ bar

Adsorption		Desorption				
P_{gauge} (bar)	P (bar)	v_{N_2} (sccs)	F_{N_2} (mol/s)	β (K/s)	T_P (K)	$n_{CO_2,ads}$ (mol)

13. Do the peak temperatures change with initial CO₂ loading? What does that suggest about the assumption of first-order desorption? Is CO₂ desorption from the zeolite first order?

14. Make measurements at the same conditions as the last experiment (same P , P_{N_2} , β) but change the N₂ volumetric flow rate v_{N_2} . Record the data in Table 1. Compare the values of T_p and $n_{CO_2,ads}$. Are they different? Should they be? Explain

Experiment #2 Estimating heat of desorption by varying heating rate

Questions to answer before starting experiment

1. If the desorption peak shifts to higher temperature at higher heating rate, does that mean the zeolite has changed?
2. Should the integrated amount of CO₂ desorbed depend strongly on heating rate if the same initial adsorption pressure is used? Why or why not?

Experimental procedure

1. Using the same procedure as for experiment #1, make measurements at four heating rates β . Keep the amount adsorbed the same by adsorbing at 7.0 bar pressure. Record the CO₂ pressure P_{CO_2} and the N₂ molar flow rate F_{N_2} .
2. Record in Table 2 the heating rate, the peak temperature T_p , and the maximum desorption rate $r_{CO_2,des,max}$ for each experiment.
3. Calculate $\frac{1}{T_p}$, $\ln\left(\frac{\beta}{T_p^2}\right)$, and $\ln(r_{CO_2,des,max})$ and enter these values in Table 2.

Table 2 Variation in heating rate $P_{CO_2} = \text{-----}$ $F_{N_2} = \text{-----}$

β (K/s)	T_P (K)	$\frac{1}{T_P}$ (K ⁻¹)	$r_{CO_2,des,max}$ (mol/s)	$\ln\left(\frac{\beta}{T_P^2}\right)$	$\ln(r_{CO_2,des,max})$

4. Plot the rate of desorption versus temperature for the four experiments. Integrate the area under curve $\int y_{CO_2,out} F_{N_2} dT/\beta$ to determine the amount adsorbed. Is it the same for each heating rate?

5. Calculate the desorption activation energy E_{des} in two ways and compare the results.

$$\ln\left(\frac{\beta}{T_P^2}\right) = -\frac{E_{des}}{R}\left(\frac{1}{T_P}\right) + K_1 \quad \text{where } K_1 \text{ is a constant}$$

Plot $\ln\left(\frac{\beta}{T_P^2}\right)$ versus $\frac{1}{T_P}$. The slope is $-\frac{E_{des}}{R}$

$$E_{des} = \text{-----} +/-$$

$$\ln(r_{des,max}) = -\frac{E_{des}}{R} \left(\frac{1}{T_p} \right) + K_2 \quad \text{where } K_2 \text{ is a constant}$$

Plot $\ln(r_{des,max})$ versus $\frac{1}{T_p}$

$$E_{des} = \text{-----} +/ -$$

Compare the desorption activation energies, where are equal to the heats of desorption, from the two plots.

Questions to answer

1. Why does the CO₂ desorption peak temperature shift to higher temperatures when the heating rate increases?
2. Is the heat of adsorption the same at high and low loadings?

3. Suppose this zeolite is used in a temperature-swing adsorption process for CO₂ capture. Based on your results, what operating changes would increase CO₂ working capacity? What changes would reduce the energy required for regeneration?
4. Where might these measurements have errors?
5. What safety precautions would you take to conduct this experiment in the laboratory?